Zero & Negative Exponents

What does a power of 0 mean? Or a negative power? These rules let the laws of exponents work for every integer — and they appear in lots of 1-mark questions.

Three rules to remember

1. **Zero exponent:** any non-zero number to the power 0 equals **1**. $n^0 = 1$ ($n \neq 0$). e.g. $7^0 = 1$, $(-5)^0 = 1$.
2. **Negative exponent = reciprocal:** $n^{-a} = 1/n^a$. e.g. $3^{-2} = 1/3^2 = 1/9$. Also $1/n^{-a} = n^a$.
3. **Power of a fraction:** $(a/b)^n = a^n/b^n$, and a negative power **flips** the fraction: $(a/b)^{-n} = (b/a)^n$.

Big idea: a negative exponent does NOT make a number negative. It means "take the reciprocal." $3^{-2} = 1/9$, which is positive.

All five laws still apply with zero and negative exponents. For example $3^5 \div 3^{-6} = 3^{5-(-6)} = 3^{11}$.

Why $n^0 = 1$? Because $n^a \div n^a = n^{a-a} = n^0$, and any number divided by itself is 1.

Where marks slip away

Trap 1 — Negative power ≠ negative number. $2^{-3} = 1/8$, not -8 and not -6.

Trap 2 — $n^0 = 1$, not 0. Any non-zero base to the power 0 is 1.

Trap 3 — Forgetting to flip the fraction. $(2/5)^{-2} = (5/2)^2 = 25/4$. Flip first, then square.

Trap 4 — Subtracting a negative exponent. $3^5 \div 3^{-6} = 3^{5-(-6)} = 3^{11}$, not 3^{-1}.

Slip 5 — Sign with fractions. $(-3/4)^{-3} = (-4/3)^3 = -64/27$ (odd power keeps the minus sign).

Model answers — write them like this

Example A. Find the value of 3^{-2}.

Step 1: Negative exponent → reciprocal: $3^{-2} = 1/3^2$.

Step 2: $3^2 = 9$, so $3^{-2} = 1/9$.

$\boxed{\therefore\ 3^{-2} = 1/9}$

Example B. Find the value of $1/4^{-2}$.

Step 1: $1/n^{-a} = n^a$, so $1/4^{-2} = 4^2$.

Step 2: $4^2 = 16$.

$\boxed{\therefore\ 1/4^{-2} = 16}$

Example C. Find the value of $(2/5)^{-2}$.

Step 1: Negative power flips the fraction: $(2/5)^{-2} = (5/2)^2$.

Step 2: $(5/2)^2 = 25/4$.

$\boxed{\therefore\ (2/5)^{-2} = 25/4}$

Example D. Simplify $3^5 \div 3^{-6}$.

Step 1: Same base, dividing → subtract exponents: $5 - (-6)$.

Step 2: $5 - (-6) = 5 + 6 = 11$.

$\boxed{\therefore\ 3^5 \div 3^{-6} = 3^{11}}$

Example E. Express 16^{-2} as a power with base 2.

Step 1: Write 16 as a power of 2: $16 = 2^4$.

Step 2: $16^{-2} = (2^4)^{-2} = 2^{4\times(-2)} = 2^{-8}$.

$\boxed{\therefore\ 16^{-2} = 2^{-8}}$

Example F. Simplify $2^{-4} \times 2^7$ and write with a positive exponent.

Step 1: Same base, multiply → add exponents: $-4 + 7 = 3$.

Step 2: So the answer is $2^3 = 8$.

$\boxed{\therefore\ 2^{-4} \times 2^7 = 2^3 = 8}$

Next: try the worksheet, then check the answer key.